Hammad Ur Rahman

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Education:

Koç University (2019 to 2025)
Ph.D. in Biomedical Science and Engineering
International Islamic University Islamabad(2013-2017)
Bachelors in Electronics Engineer
F.G Sir Syed College The Mall, Rawalpindi(2010-2012)
HSSC Pre-Engineering

Experience:

Assistant Professor at College of Electrical and Mechanical Engineering, NUST, Islamabad (2024 to present)
Research assistant at Manufacturing & Automation Research center (MARC) Koç

Projects:

Advanced Research for Unmanned Air Vehicles, Baykar Savunma, 2020-2021
Novel Sensor Development for Manufacturing, Ford Otosan, 2020-2021.
Artificial Heart Pump System, TÜBİTAK, 2020-2023

Publications:

1. **Effective image processing-based technique for frost detection and quantification in domestic refrigerators.**
   Hammad ur Rahman, Hassan Akbar, Anjum Naeem Malik, Tahir Nawaz, Ismail Lazoglu
   International Journal of Refrigeration
   https://doi.org/10.1016/j.ijrefrig.2024.01.026
2. **Ice growth detection and the de-icing using dual functional capacitive sensor**
   Hammad Ur Rahman, Anjum Naeem Malik and Ismail Lazoglu
   Measurement Journal
   https://10.1016/j.measurement.2024.115208
3. **A centralized frost detection and estimation scheme for Internet-connected domestic refrigerators**
   Hammad Ur Rahman, MU Mehmood, I Lazoglu
   International Journal of Refrigeration 169, 194-203
   https://10.1016/j.ijrefrig.2024.10.032
4. **Machining process monitoring using an infrared sensor**
   W Akhtar, **Hammad Ur Rahman**, I Lazoglu
   Journal of Manufacturing Processes 131, 2400-2410
   https://10.1016/j.jmapro.2024.10.063
5. **Frost detection and defrost control: Emerging trends, methods, and experimental evaluations**
   H Hassan, H Akbar, AN Malik, T Nawaz, H Elahi, **Hammad ur Rahman**, I Lazoglu
   International Journal of Refrigeration
   https://10.1016/j.ijrefrig.2025.08.005
6. **Frost-EffNet: A deep learning model for frost detection in refrigeration systems**
   MU Mehmood, HU Rahman, DB Desdemir, I Lazoglu

International Journal of Refrigeration
https://10.1016/j.ijrefrig.2025.09.029

7. **Photogrammetry, additive manufacturing, and sensorization for archaeological ceramic loss compensation**
   SU Bishop, M Arshad, I Lazoğlu, M Harpster, **Hammad Ur Rehman**
   Journal of Cultural Heritage 76, 148-156
   https://10.1016/j.culher.2025.09.012

8 **Frost Detection and Thickness Estimation using a Transformer-based Semantic Segmentation Model**
   H Hassan, H Akbar, AN Malik, T Nawaz, H Elahi, Hammad ur Rahman, I Lazoglu
   International Journal of Refrigeration, 106899
   https://10.1016/j.ijrefrig.2026.106899

9. **Real-time physiological environment emulation for the Istanbul heart ventricular assist device via acausal cardiovascular modeling**
   Hammad Ur Rahman, K Mehmood, F Abdulhamid, I Lazoglu, V Bakuy, ...
   Artificial Organs 49 (3), 410-419
   https://doi.org/10.1111/aor.14903Digital Object Identifier (DOI)

10 **A Multimodal CNN–BiGRU Approach to Ventricular Tachycardia Alarm Validation**
   V Khurram, M Azeem, M Javed, AN Malik, H Elahi, Hammad ur Rahman
   2026 International Conference on Robotics and Automation in Industry (ICRAI)
   https://10.1109/ICRAI70912.2026.11551960

11 **EFFECTS OF MILLING TYPE ON CUTTING FORCES AND SURFACE FINISH IN ROBOTIC MACHINING**
   IL Mehmet Ali Erhun, Mohammadreza Chehrehzad, Ulas Agdogan, Hammad Ur Rahman
   Proceedings of 12th UTIS International Congress on Machining 12, 278-287

Patents:

1. **An Ice Sensor**.
   PCT/TR2021/050595
   Inventors: ***Hammad ur Rahman***, Ismail Lazoglu

2. **Infrared based low-cost in situ monitoring sensor and method for workpiece and tool condition monitoring during machining**
   PCT/TR2023/050598
   Inventors: Waseem Akhtar, ***Hammad ur Rahman***, Ismail Lazoglu

3. **Metal forming apparatus and method of determining Draw-in**
   2022/014803
   Inventors: Berkay Demiryülek, ***Hammad Ur Rahman***, Anjum Naeem Malik, Vügar Kerimoğlu, Oğuz Yasin, Ismail Lazoğlu

Courses taught in CEME, NUST (2024 onwards)

- Solid Modeling
- Linear Control System
- Modeling and simulation
- Electronic Devices and Circuits
- Introduction to Biomedical Engineering

Teaching assistant at Koç University (2019- 2024)

- Statistics.
- Physics Laboratory.
- Dynamics.
- Intro to Mechanical Engineering Device
- Mechatronics

OSP Engineer at GCS Pvt Ltd.(2017-2019)
- RFP & technical solution designing of corporate clients.
- Provisioning and commissioning of optical fiber optics and CCTV.
- FTTH & GPON network designing and implementations.
- Maintain records of Optical Fiber cores utilization.

Skills:

Matlab, Siemens NX, Proteus, Comsol Multiphysics, Arduino, Integrated Circuit Design, Lab view, Multisim, adobe photoshop, conducting a literature review, data analysis, data processing, research protocols, critical thinking, analytic problem-solving.

Courses:

Mechanics of Microsensors, Biomedical signal processing, Sustainable Energy, Mechanics and material in medicine, Biological fluids mechanics, Biomimetics, Micro & Nano Fabrication, Metal forming, Artificial intelligence in metallurgy, Refinery technologies, Computer Integrated Manufacturing, and Automation, Manufacture.

Achievements:

Secured 1st position in Robomove, TECTICS’16 IQRA University.
Secured 2nd position in Roborumble, Air TECH’16 AIR University.
Secured 1st position in Robomaze, NEO’17 GIKI University.
Secured 1st position in line following, FETEX’16 IIUI University.